

One-month strawberry-rich anthocyanin supplementation ameliorates cardiovascular risk, oxidative stress markers and platelet activation in humans.

Strawberries are an important fruit in the Mediterranean diet because of their high content of essential nutrients and beneficial phytochemicals, which seem to exert beneficial effects in human health. Healthy volunteers were supplemented daily with 500 g of strawberries for 1 month. Plasma lipid profile, circulating and cellular markers of antioxidant status, oxidative stress and platelet function were evaluated at baseline, after 30 days of strawberry consumption and 15 days after the end of the study. A high concentration of vitamin C and anthocyanins was found in the fruits. Strawberry consumption beneficially influenced the lipid profile by significantly reducing total cholesterol, low-density lipoprotein cholesterol and triglycerides levels (-8.78%, -13.72% and -20.80%, respectively; $P < .05$) compared with baseline period, while high-density lipoprotein cholesterol remained unchanged. Strawberry supplementation also significantly decreased serum malondialdehyde, urinary 8-OHdG and isoprostanes levels (-31.40%, -29.67%, -27.90%, respectively; $P < .05$). All the parameters returned to baseline values after the washout period. A significant increase in plasma total antioxidant capacity measured by both ferric reducing ability of plasma and oxygen radical absorbance capacity assays and vitamin C levels (+24.97%, +41.18%, +41.36%, respectively; $P < .05$) was observed after strawberry consumption. Moreover, the spontaneous and oxidative hemolysis were significantly reduced (-31.7% and -39.03%, respectively; $P < .05$), compared to the baseline point, which remained stable after the washout period. Finally, strawberry intake significantly decreased ($P < .05$) the number of activated platelets, compared to both baseline and washout values. Strawberries consumption improves plasma lipids profile, biomarkers of antioxidant status, antihemolytic defenses and platelet function in healthy subjects, encouraging further evaluation on a population with higher cardiovascular disease risk. Copyright © 2014 Elsevier Inc. All rights reserved.